

Final Report

Kofi Asante Ampim Darkoh

July 24, 2025

Generative Modeling by Estimating Gradients of the Data Distribution

My time working with the wonderful instructors from AI-EDGE Institute has been very eye opening and an experience I would advocate for rising juniors in the upcoming year to partake in due to the accommodating nature of the teaching environment.

I came into this experience with some knowledge of uploading artificial intelligence and machine learning models on edge devices such as Raspberry Pi Pico and google coral development boards. Though that knowledge was very useful, I still lacked the advanced tools to build such models on my own, but this is no longer the case due to my current expertise in artificial intelligence fields such as generative models for images and texts, artificial intelligence for security design and distributed AI and federated learning.

The first half of the program spanning weeks 1 through 4 ensured that my fundamental understanding of machine learning and artificial intelligence concepts was solidified. Learning about wireless networks was a groundbreaking moment as I was afforded the opportunity to learn the integral role of networking in holding models together. Weeks 5 through 8 focused on more advanced concepts such as integrating artificial intelligence for wireless designs, having text generation models, security in artificial intelligence and finishing up the program by having team projects or individual reports; the latter of which I was assigned due to my online status. I chose to focus this report on the topic *Generative Modeling by Estimating Gradients of the Data Distribution*, an article by Yang Song, a professor at Stanford University; the article focuses on generative modeling, score-based to be specific and the benefits of such a model over avoiding adversarial training and likelihood computation both of which are traditional models and lack the ability to generate their own samples using techniques such as stochastic differential equations.

The first concept the article discusses is the score function and score matching, where we take the gradient for the log-density of the data provided. The score function is easy to use to make a model, and this is because unlike other methods such as the probability density we are not required to provide normalization thus giving a free ticket to make a model. Although Variational Autoencoders (VAEs) and normalization flows, traditional likelihood-based models, learn compressed data and generate new data based on that, we can do this with the score function and not have to do any normalization on the data, which on a small dataset may seem insignificant but on a larger dataset where time is essential or time lost could mean money lost, we cannot afford to spend time on unessential processes.

The scoring function also provides use with an easier architecture to work with which is very flexible and this is due to the scoring function being independent of the normalization constant and as such allowing score-based models the ability to use arbitrary neural networks without any issues.

When we use the score function and score matching system, we aim to reduce the Fisher divergence which may exist between the true score and model score. We begin without a score matching value hence use the denoising score matching (DSM) to make an approximate score matching value to create noise in the data.

Overall, the score function and score matching method enable us to solve a lot of problems. For example, having datasets where scores are estimated widely off the mark can be solved by adding noise to the data, Langevin dynamics. Another issue we can solve with this method is the curse of dimensionality, where data requires being regulated; we change this by spreading scores on random locations to improve the efficiency of our model.

Next, we have Langevin Dynamics for Sampling which is a Markov Chain Monte Carlo (MCMC) sampling method that uses the score function to refine random noise a dataset. Langevin Dynamics is the most integral aspect in score-based generative models as it allows us to transform random noise into meaningful samples using iterative refinement. The equation we use in this method is $x_{t+1} = x_t + \epsilon \nabla \log p(x_t) + \sqrt{2\epsilon} z_t$ (where $z_t \sim N(0, I)$). Here, we move portions of the model to higher density regions and add noise to escape local maxima.

Langevin Dynamics ensures that the model produces samples from the target distribution and this is due to concepts such as Fokker-Planck Equation which corresponds to a discretization of the stochastic differential equation, the detailed balance which is a combination of drift and diffusion to satisfy the balance condition that all MCMC methods are required to have to converge to the correct distribution and finally ergodicity which allows the sampler explore the regions of probability space proportional to their density.

One may assume that this is very great, however all the above are pretty much guaranteed if the stars align and as such for practical conditions we may have to look at discretization error – finite step sizes to introduce bias in samples or mixing time and finally, curse of dimensionality, where convergence becomes slower in high-dimensional spaces.

This may beg the question, would these practical conditions allow for the Langevin Dynamics method to perform strongly, and the answer is yes. This is because it incorporates second-order data through a preconditioning matrix and helps move through difficult probability landscapes. Another feature of it is the Metropolis-Adjusted Algorithm which guarantees accurate sampling at the cost of additional computation. We can also enable mini-batch training for large datasets and use noisy gradient estimations while maintaining convergence to provide a more suitable ground for the Langevin Dynamics method to thrive.

The method has issues such as slow mixing in high dimensions, occasional cases of being sensitive to step size selection, computational expense for large models and faces difficulty with regards to dealing with discrete variables.

We are then shown a sneak peek into Stochastic Differential Equations using the equation $dx = f(x, t)dt + g(t)dw$, we are able to have extended formulations. On the everyday scale, we would be using SDE discretization schemes such as Euler-Maruyama, Milstein or Adaptive step-size approaches to arrive at a consensus solution. Another approach we could consider is one that involves achieving numerical stability to ensure that the figures we arrive at are consistent and we do this through error control, stiffness handling and step size adaptation. Now, in the instance where the above two approaches may not be suitable for what we intend to do, we can use the hybrid solvers approach which is much more secure than the former and uses advanced techniques such as splitting methods and predictor-corrector to manage the dataset.

We can also look much deeper into SDEs and find that there are much more advanced aspects of this which includes an ability to control what is defined within the confines of what we are working with, and this can be done using image boundaries and manifold learning. We can also use latent ODEs, implicit networks and neural operators for the fast sampling of methods, an approach which makes the whole process of using the Stochastic Differential Equations relatively useable and a widely used method for generating models.

Using Stochastic Differential Equations in score-based generative modeling, we are allowed to generalize discrete noise perturbations into a continuous-time process and connect to diffusion models and thermodynamics thereby allowing cross-disciplinary insights. SDEs also provide exact likelihood computations by providing probability flow ODEs that can tract density estimates. They also perform inverse problem solving naturally integrating with Bayesian inference for inpainting and MRI reconstruction.

However, SDEs also provide complex implementation which require careful tuning of diffusion coefficients and are sensitive to step-size selection and noise schedules which can be time consuming. Another problem with SDEs is that they do not allow for the storage of intermediate states for ODE-based likelihoods that are not allowed for high-resolution data.

Next on the list is Annealed Langevin Dynamics which is also a powerful sampling technique that progressively refines samples using some sequence of noise levels. By combining principles from non-equilibrium thermodynamics and stochastic calculus, ALD enables efficient exploration of complex, high-dimensional data distributions. We can interpret this through thermodynamic annealing which is a process that resembles temperature cooling in physical systems, where the effective distribution transitions smoothly from a high-noise state to the target data distribution.

Also, under mild conditions, ALD achieves exponential convergence in total variation distance, with finite-time error bounds and provably optimal cooling schedules.

Now, to enhance performance, modern implementations incorporate through adaptive transition schemes using techniques such as overlapping noise windows and momentum-based updates and hybrid sampling which combine ALD with Hamiltonian Monte Carlo or slice sampling boosts efficiency.

Parallelization is also another approach to look at where multi-chain ensembles and distributed sampling reduce computational overhead effectively leading to a stable and consistent sampling for the dataset we work with.

Some of the strengths of the ALD are that it produces high-quality samples by systematically traversing noise levels, is compatible with both discrete and continuous noise schedules and is theoretically grounded with convergence guarantees but it comes with its own limitations such as being computationally intensive due to an iterative refinement and sensitive to parameter choices such as cooling rate.

ALD bridges theoretical rigor with practical utility, making it indispensable for score-based generative modeling. Future work may focus on adaptive cooling schedules and hybrid methods to further accelerate sampling while maintaining quality.

To conclude, I believe that the framework of score-based generative modeling represents a significant leap forward in machine learning, offering a unified approach to sample generation, density estimation, and inverse problem solving. At its core, the methodology's reliance on **score functions**—gradients of log-probability densities—rather than explicit density modeling circumvents the need for restrictive architectures or adversarial training, which has long been a bottleneck in generative AI. This shift not only simplifies the theoretical underpinnings but also enhances practical flexibility, allowing models to be trained on high-dimensional data without the instability plaguing GANs or the normalization constraints of VAEs.

One of the most compelling aspects of this paradigm is its **theoretical elegance**. By framing the generative process through stochastic differential equations (SDEs), the method bridges discrete noise perturbation (e.g., annealed Langevin dynamics) with continuous-time diffusion, offering a cohesive mathematical narrative. The reverse-time SDE formulation elegantly demonstrates how noise corruption can be systematically undone through score-guided trajectories.

Looking ahead, the field must address **scalability and accessibility**. Recent efforts in distillation and hybrid architectures (e.g., combining SDEs with latent-space models) hint at solutions, but the computational overhead still limits widespread adoption. Additionally, the theoretical depth of SDEs may deter practitioners, calling for more intuitive abstractions and tooling.

Despite these challenges, the framework's mathematical coherence and empirical success position it as a cornerstone of generative AI's future—one that harmonizes theory and practice in ways few other methods achieve.

Finally, I believe that score-based generative modeling is a testament to the power of gradient-based learning. Its innovations redefine what's possible in generative tasks, though practical barriers remind us that no method is a panacea. As research tackles these limitations, the framework's influence will likely expand, potentially reshaping not just generative AI but also adjacent fields like optimization and computational physics. Now, are

there other ways that we can have better results in the field of generating models? Yes. But for a sure proof method of generating high quality models, the score-based system appears to be the best.